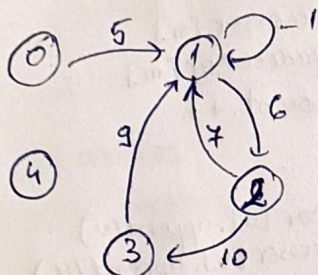


Lab1 - Graphs

24.02.2026

diagram:



graph.txt:

mr. vertices	5	6	mr. edges
0	1	5	
1	1	-1	
1	2	6	
2	3	10	
2	1	7	
3	1	9	

dictionaries:

D	precessor
0	
1	[0,1,2,3]
2	[1]
3	[2]
4	
5	
6	

↓
6 elem.

↓
"WHAT COMES TO ME"

D	successor
0	[1]
1	[1,2]
2	[1,3]
3	[1]
4	
5	
6	

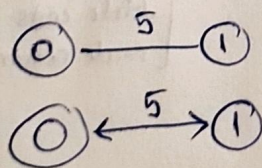
↓
6 elem.

↓
"OUT"

D	cost
0	<0,1> = 5
1	<1,1> = -1
2	<1,2> = 7
3	<2,3> = 10
4	<2,1> = 7
5	<3,1> = 9

↓
6 elem.

undirected graph:



→ inside : $\begin{cases} 0 & 1 & 5 \\ 1 & 0 & 5 \end{cases}$

class: modify

① ADD (6) ← (5)

graph-modified.txt

6 6
0 1 5
1 1 -1
1 2 6
2 3 10
2 1 7
3 1 9

ADD : { empty successor[m]
empty predecessor[m]
vertex count.++
mode m

ADD : { successor[u].append(v)
predecessor[v].append(u)
cost[(u,v)] = c
edge u → v,
cost c
edge count ++

<6,4> = 5

DELETE : { remove v from successor[u]
remove u from predecessor[v]
remove cost[(u,v)]
edge u → v
edge count --

② DELETE a mode

② (mode)

- "how many things you delete from the cost, you delete from p and s"
- delete all apairitions of ②, but count them as well!

D precessor
1, 2
2, 3
2, 1

D successor
2, 1
3, 2
1, 2

→ after deletion?

DELETE : { remove all edges from m
remove all edges to m
delete successor[m]
delete predecessor[m]
delete costs involving m
mode count --
mode m

an edge

(0-1)

D precessor
0, 1

D successor
1, 0

Class/Struct (for the graph)

- Nbr. of vertices
- D pre
- D suc
- D cost

③ CHANGE

$$(3,4), (2,3) = 1$$

→ cost $\exists ? / \nexists ?$

Menu

- 1) get vertices
- 2) parse vertices
- !
- ! external function